

- 1) A network shows an open circuit voltage of 9.0V and a short-circuit current of 1.5A at its terminals. Find R_{th} , R_N and the terminal voltage for a 4Ω load.

⇒ open circuit voltage (V_{th}) = 9V.
 Short circuit current $I_N = I_{sc} = 1.5A$
 Load resistance (R_L) = 4Ω.

From thevenin-norton's circuit, $R_{th} = \frac{V_{th}}{I_{sc}} = \frac{9}{1.5} = 6\Omega$.
 $\therefore R_N = R_{th} = 6\Omega$.

Terminal voltage for 4Ω load,

$$V_L = V_{th} \cdot \frac{R_L}{R_{th} + R_L} = \frac{9 \times 4}{6 + 4} = \frac{36}{10} = 3.6V.$$

The formula is coming from thevenin's equivalent circuit.

- 2) A 100mH inductor and a 47μF capacitor are placed in series. Find the frequency at which their reactance are equal.

⇒ $L = 100mH = 100 \times 10^{-3} H$
 $C = 47\mu F = 47 \times 10^{-6} F$.

we know that,

$$X_L = X_C$$

$$\omega L = \frac{1}{\omega C}$$

$$\omega^2 = \frac{1}{LC}$$

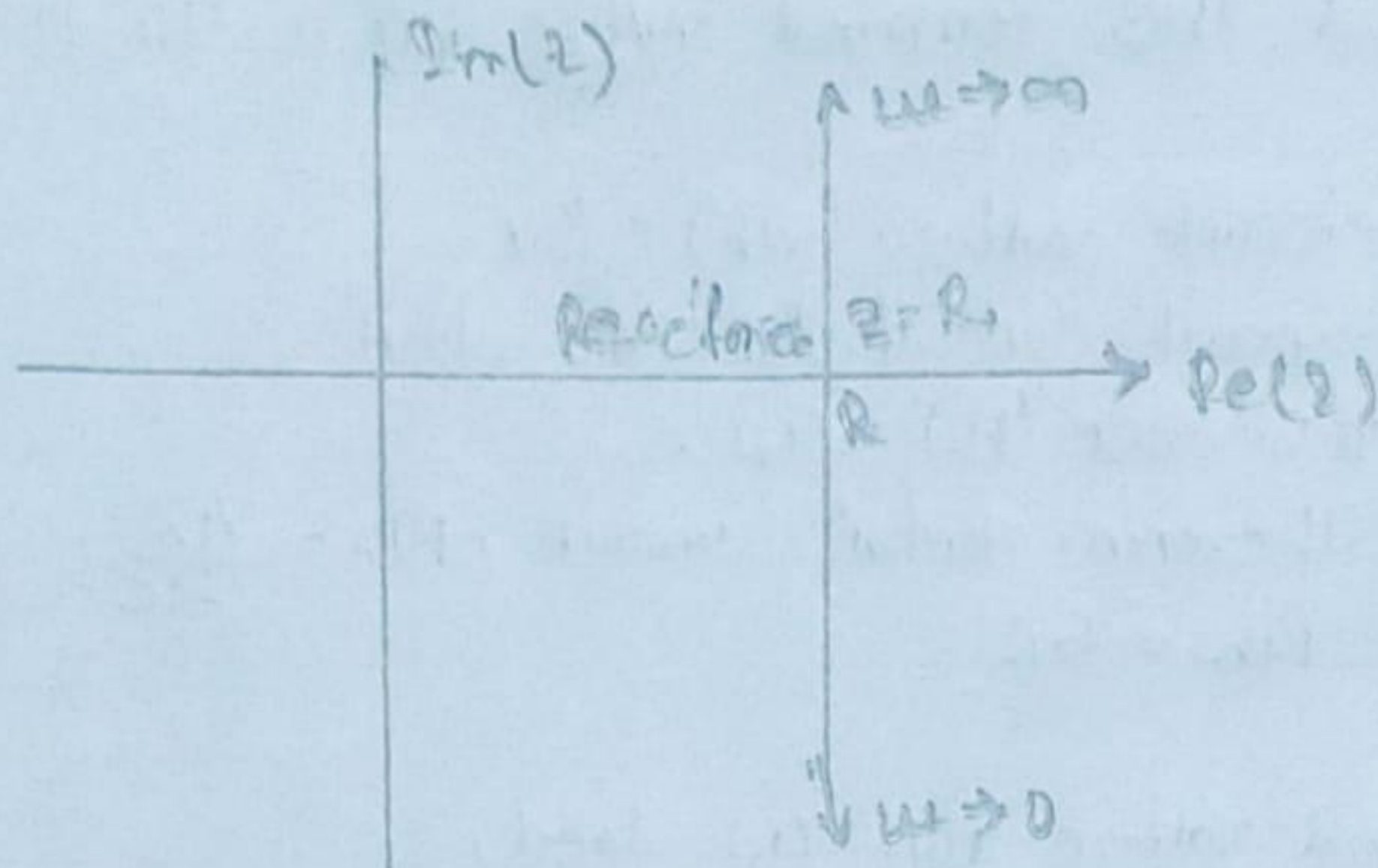
$$\omega = \frac{1}{\sqrt{LC}}$$

$$\omega = \frac{1}{\sqrt{100 \times 10^{-3} \times 47 \times 10^{-6}}}$$

$$= \frac{10^4}{\sqrt{4700}} = 461.9 \text{ rad/s}$$

Impedance locus,

the locus is a vertical straight line at $\text{Re}(Z) = R$



Name the frequency is,

$$f = \frac{\omega}{2\pi} = \frac{461.9}{2\pi} = 73.55 \text{ Hz.}$$

3) An inductor voltage waveform is $+20\text{V}$ for 3ms and $-V_2$ for the remaining 2ms of a 5ms period. Find V_2 .

2) Given,

Total period $= 5\text{ms}$

At Steady state operation,

$$\int_0^T V_L(t) dt = 0$$

$\therefore$ Positive and negative volt-seconds must be equal,

Apply volt-second balance,

$$20(3 \times 10^{-3}) + (-V_2)(2 \times 10^{-3}) = 0$$

$$\Rightarrow 60 \times 10^{-3} - 2V_2 \times 10^{-3} = 0$$

$$\Rightarrow 60 = 2V_2$$

$$\Rightarrow V_2 = 30\text{V}$$

$\therefore$ waveform of Inductor $+20\text{V}$ for 3ms and -30V for 2ms .

4) Sketch the impedance locus of a series RLC circuit as ω is swept from 0 to ∞ , and identify the sketch at which the locus crosses the real axis.

2) For a series RLC circuit,

$$Z = R + jX_L - jX_C$$

$$\because X_L = \omega L \text{ and } X_C = \frac{1}{\omega C}$$

$$\therefore Z = R + j(\omega L - \frac{1}{\omega C})$$

$$\text{Re}(Z) = R \text{ and } \text{Im}(Z) = \omega L - \frac{1}{\omega C}$$

At $\omega = 0$, $X_L = 0$, but $X_C \rightarrow \infty$

$$\therefore \text{Im}(Z) \rightarrow -\infty$$

So, the locus starts at $R - j\infty$

At resonance,

the locus crosses the real axis when $X_L = X_C$

$$\therefore \omega_0 L = \frac{1}{\omega_0 C} \Rightarrow \omega_0 = \frac{1}{\sqrt{LC}}$$

At this point, $X_L - X_C = 0 \Rightarrow$ So, $Z = R$

At $\omega \rightarrow \infty$, $X_L \rightarrow \infty$ and $X_C \rightarrow 0 \therefore \text{Im}(Z) \rightarrow +\infty$

So the locus goes toward $R + j\infty$

5) Using Euler's Formula, express $V(t) = 20 \cos(314t + 30^\circ)$ as a phasor, and state the frequency in Hz.

$$\Rightarrow \text{Given, } V(t) = 20 \cos(314t + 30^\circ)$$

Standard sinusoidal form, $V(t) = V_m \cos(\omega t + \phi)$

Here, $V_m = 20$, $\omega = 314 \text{ rad/s}$, $\phi = 30^\circ$

$\therefore$ Phasor is $20 \angle 30^\circ \text{ V} = V'$

$$V_{\text{rms}} = \frac{20}{\sqrt{2}} = 14.14 \text{ V then, } V' = 14.14 \angle 30^\circ \text{ V rms}$$

We know,

$$\omega = 2\pi f \Rightarrow f = \frac{314}{2\pi} = 49.97 \text{ Hz} \approx 50 \text{ Hz}$$

Teacher's Signature